

Octo · arXiv 2405.12213 · RSS 2024

Visionary

11+3

Teach Octo / run $\pi_{0.5}/N1.7$

Durable: compositional tokens + chunk head

Oct 14 paper club · VLA Architectures · Oct 16 BEHAVIOR nearby

Spine: open OXE GRP + diffusion chunks · underspecified horizons · NOT 2026 winner weights

Octo Model Team · Berkeley / Stanford / CMU / DeepMind · HF rail-berkeley/octo-*

Visionary map · 2026 BEHAVIOR (no Discord/Drive)

1 Teach / run

Teach Octo lessons
Run $\pi_{0.5}$ / N1.7 for Oct 16
Octo = ancestor, not baseline

2 Steal virtues

Shared FT HPs
Adapter / new-I/O tests
App. E negative results

3 Control defaults

Chunk + receding (not TE)
Backbone-once + light head
Absolute gripper alignment

4 Durable idea

Compositional multi-robot
tokenization +
generative chunk head

5 Ablate

FM vs DDPM on same VLM
Mixture card > romantic scale
Wrist/lang coverage first

6 Honest

72% short-lab $\neq$ BEHAVIOR q
Cite open GRP + chunk FT
Not household SOTA

Genealogy · CLEAR club line



Pivot: open ViT + diffusion-chunk GRP on 800k OXE with block-wise adapters

Octo = infrastructure history + eval baseline · NOT 2026 winner weights

Priors: OXE/RT-X · DP · ACT · ViT · T5 · DDPM · HER goals · RT-1/RoboCat · Bridge

Subsequent CLEAR: OpenVLA · π_0 (2410.24164) · OpenPI · $\pi_{0.5}$ /RTC · BEHAVIOR'25-'26

Club line

**ACT || DP → Octo (open OXE GRP)
→ π_0 / OpenVLA → OpenPI / BEHAVIOR**

π_0 (CLEAR) beats Octo-93M on hard dexterous tasks with VLM+FM.
Octo = infrastructure history + eval baseline — not 2026 winner weights.

Prefer $\pi_{0.5}$ / N1.7 for BEHAVIOR · keep Octo for open FT ancestor + mixture lessons

Durable Octo idea

Compositional multi-robot tokenization

+

Generative action chunk head

Generative family will keep changing (DDPM → FM → hybrids)
Open multi-embodiment I/O remains the product requirement

Skip Discord/Drive · stay paper- and sibling-pack grounded

FT recipe card · operationalizable (rare honesty)

Data

~100 target demos

Steps

50k · cosine + warmup

Update

Full model (beat freeze)

Compute

~5 h · A5000 24 GB

Adapters

New cam / F/T / joint / 14-DoF

Result

Table I avg 72% vs 20%/15%

Domains: Insertion* (F/T) · Coffee · Baking · Pick-Up† · Coke · Bimanual† (*new obs †new action)

Underspecified horizons · motion FLAG

Main text: “several” future actions · only concrete: bimanual pred 64 / exec 12

System	Chunk object	Generative	Ensemble?	Typical commit
ACT	k×14 joints	CVAE	Yes TE	ensembled @ 50 Hz
Diffusion Policy	Tp actions	DDPM/DDIM	warm / Ta	Ta≈8 @ ~10 Hz
Octo (this)	action chunk	DDPM readout	No (paper)	receding prefix
π_0 / $\pi_{0.5}$	H actions	Flow matching	No	open-loop K @ 20-50
BEHAVIOR'25	H≈30-32	FM + corr.	soft inpaint	~30 Hz receding

Empiricist: always log (H_pred, H_exec, Hz, diff_steps) · commit_seconds = H_exec/Hz · TE not dead globally

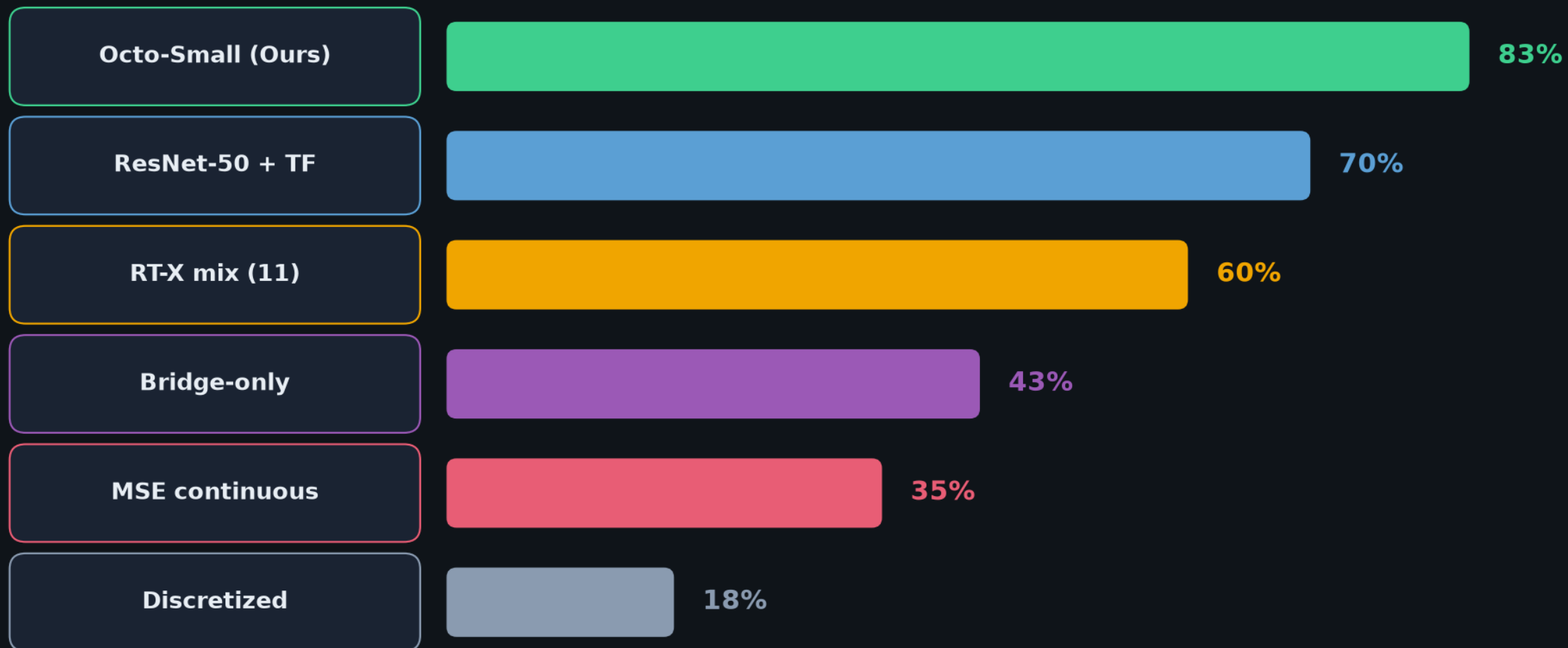
Generalization honesty · Table VII WidowX



Cite 5% novel skill whenever someone says “generalist zero-shot”

Objects transfer; environments degrade; skills near floor — prevents foundation-model overclaim

Ablation board · Table II WidowX Octo-Small



Diffusion $\gg$ MSE/discrete · mix width 43 $\rightarrow$ 60 $\rightarrow$ 83 · ViT-first helps at pretrain scale

Stakeholder value · open FT ancestor, not product brain

Open usable GRP

27M / 93M HF ckpts
JAX FT scripts + OXE loaders
2024 default “try a pretrained policy”

Cost model that sells

~100 demos + ~5 h A5000
→ 72% avg vs 20% scratch
Consumer-GPU honest

NOT 2026 SOTA

No Internet VLM backbone
93M vs π_0 ~3B
 π_0 CLEAR-beats hard dexterity

Strategic ask

Keep Octo in eval harness
Ship $\pi_{0.5}$ / GR00T for BEHAVIOR'26
Oct 16 deadline nearby

Leadership one-liner

Octo proved open cross-embodiment pretrain + cheap FT works;
 π_0 proved you still need VLM scale + better generative action modeling for household dexterity.

Visionary · close

Teach Octo · run $\pi_{0.5}/N1.7$ for Oct 16

Steal: shared FT HPs · adapters · App.E negatives · chunk+receding

Durable: compositional multi-robot tokens + generative chunk head

No Discord/Drive

Q&A · keep the spine clear

Octo: An Open-Source Generalist Robot Policy

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<https://octo-models.github.io>

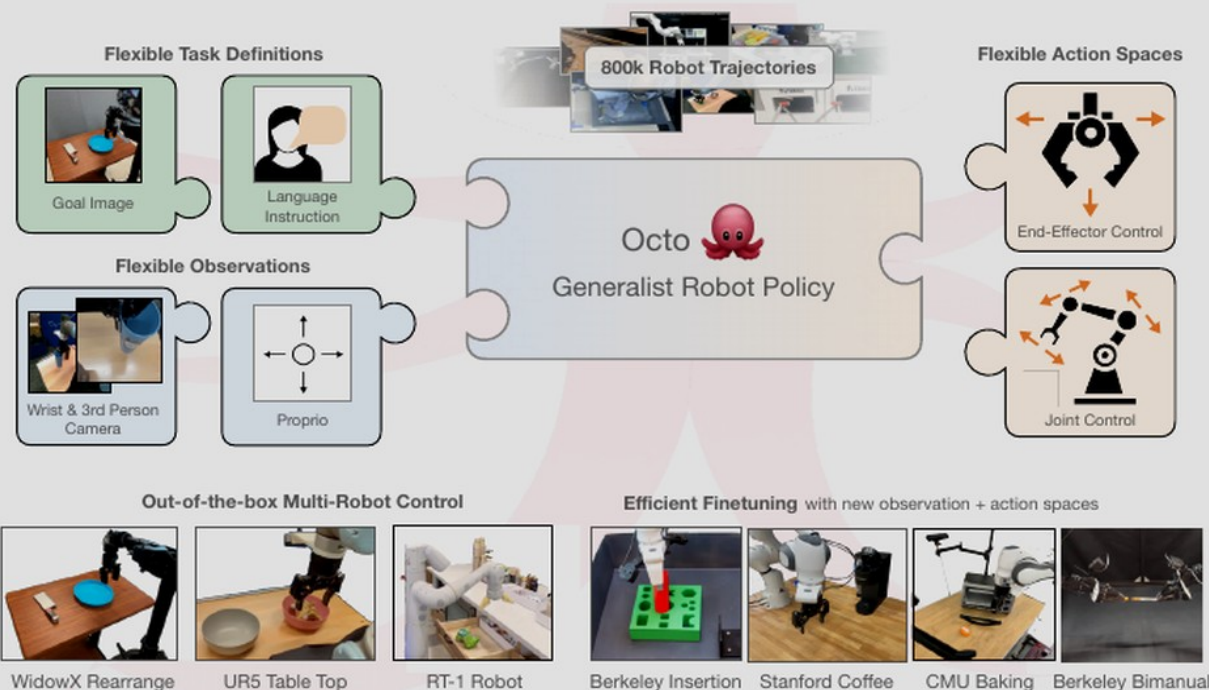


Fig. 1: We introduce Octo, an open-source, generalist policy for robotic manipulation. Octo is a transformer-based policy pretrained on 800k diverse robot episodes from the Open X-Embodiment dataset [67]. It supports flexible task and observation definitions and can be quickly finetuned to new observation and action spaces.

Abstract—Large policies pretrained on diverse robot datasets have the potential to transform robotic learning: instead of experiments across 9 robotic platforms, we demonstrate that Octo serves as a versatile policy initialization that can be effectively